

AutoPooper 3000: Colouring the Fill Lines

Filling the MIN / MAX grooves and letters inside the drum bowl with bright yellow epoxy, flush with the wall

When: last of all, after the drum tiles are welded and the solvent cement has fully outgassed, and before any litter goes in. Same for Drum Mark I and Mark II. The grooves are about 2mm wide and 1mm deep, so the whole job takes only 3–5 g of mixed epoxy.

Why yellow: the lines sit right at the litter surface under a film of light grey clay dust. A dusty white line looks like dust on the charcoal drum and disappears; yellow stays clearly yellow. Safety orange works as well. Because the colour goes all the way through, wear can never show dark plastic, and a flush line leaves no ledge for dust or urine.

What you need

Item	Notes
30-minute 2-part epoxy (1:1)	Yours, from the double syringe or two tubes. Not 5-minute epoxy: weaker, more brittle, yellows more.
Opaque epoxy resin pigment, bright yellow	Paste or liquid. The label must say it is for epoxy resin.
Glow powder (optional)	Strontium aluminate, green glow, fine grade. Not zinc sulphide (fades in minutes).
Isopropyl alcohol, 400–600 grit sandpaper	Cleaning and lightly scuffing the grooves so the epoxy grips.
Masking tape	Both edges of every line and round the letters.
Plastic lid, stir sticks, toothpicks	Mixing surface; toothpicks press the epoxy into the grooves.
Old plastic card (gift card)	Levels the fill on top of the tape.
Single-edge razor blades	Shaves any bump off while the epoxy is leathery.
800 grit wet-and-dry + a popsicle stick	Only for a bump that is left after full cure.
Nitrile gloves	Epoxy on skin can cause allergies over time.

Choosing the tint

Use	Don't use
Opaque epoxy resin pigment, paste or liquid, made for epoxy	Alcohol ink: transparent, so the dark groove shows through; too much spoils the cure
A small pinch of mica in the paste is fine for a sheen	Mica powder on its own: shimmery, see-through, settles
	Acrylic or any water-based paint, food colouring: clouds and weakens the epoxy

Dose: about 2–5% of the epoxy's weight, never above the maker's limit (usually about 5–6%), or the epoxy can stay soft. For two 2–3cm beads from the syringe that is about **1–2 drops of liquid or a match-head of paste**.

Test first (10 minutes, saves the drum)

- Do the whole routine once on the **drum-wall test coupon** with a short groove (or any PETG scrap with a groove cut in it).

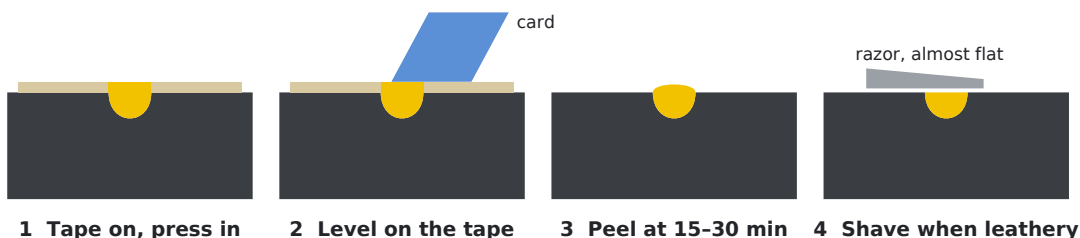
- **Cover test:** draw a line with a black marker on scrap, smear a thin layer of the tinted epoxy over it. If you can still see the line, the tint is not opaque enough: add a little more (within the limit) or switch to a paste.

Step by step

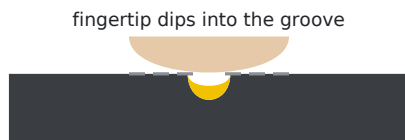
- 1 Clean.** Wipe the grooves and letters with isopropyl alcohol. Let it dry.
- 2 Scuff.** Lightly sand inside the grooves with 400–600 grit folded over a toothpick. Wipe with alcohol again.
- 3 Mask.** Run masking tape tight along both edges of every line and round the letters, right up to the groove edge. Press the edges down firmly.
- 4 Get everything ready** before opening the epoxy: lid, sticks, toothpicks, card, tint (and glow powder). In the heat you have about 10–15 minutes.
- 5 Two blobs.** Squeeze out two equal blobs of resin and hardener **side by side, not touching**. A 2–3cm bead each is plenty.
- 6 Tint one blob only.** Work the tint into one blob until there are no streaks (1–2 drops of liquid or a match-head of paste). If you use glow powder, mix it into the same blob: about a quarter of that blob's volume. Fold slowly so you don't whip in air. The 30-minute clock has not started yet.
- 7 Combine and mix.** Fold the two blobs together and mix for a full 1–2 minutes, scraping the stick clean and folding again.
- 8 Press it in.** With a toothpick, push the epoxy into the groove, moving *along* the groove so no air is trapped underneath. Fill a little above the tape. Pop any big bubbles with the toothpick (no lighter or heat gun near the PETG).
- 9 Level it.** Drag the plastic card along the line, riding on the tape on both sides. The fill is now level with the tape, a hair above the drum.
- 10 Peel the tape at 15–30 minutes,** while the epoxy is still soft: low angle, pulling away from the line. This gives clean edges.
- 11 Shave when leathery (after a few hours).** If there is a bump, lay a single-edge razor blade almost flat on the drum and shave it off. At this stage it cuts like hard cheese.
- 12 Cure about 24 hours** (check your epoxy's label) before touching it again. Give it the full cure time on the label before litter goes in.
- 13 Only if a bump is left:** wrap 800 grit wet-and-dry round a popsicle stick, wet it, and use short strokes along the line. Stop as soon as it is flush.

What it looks like

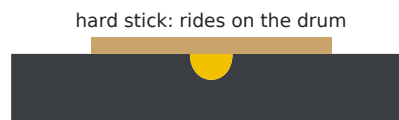
Cross-section through one groove (not to scale). Charcoal = drum wall, yellow = tinted epoxy, tan = masking tape.



Why never sand with a fingertip



WRONG: fill scooped out, drum scuffed



RIGHT: fill stays flush

A soft pad or a fingertip dips into the groove and scoops the fill out, and sanding scuffs a shiny band on the matte charcoal drum either side of the line. The card, tape and razor do the levelling; sandpaper only touches leftover bumps, on a hard stick.

Timing at a glance

When	What
0 min	Combine the two blobs and mix 1–2 minutes
0–10 min	Press into the grooves and level with the card (mix small batches)
15–30 min	Peel the tape
A few hours	Shave any bump with the razor while leathery
~24 h	Cured; sand only leftover bumps
Label's full cure	Litter can go in

If something goes wrong

Problem	Fix
Epoxy strings or goes rubbery while you work	Throw it out and mix a fresh, smaller batch.
Dark groove shows through the yellow	Tint not opaque enough. Once cured, scuff and add a thin second fill with more tint (within the limit) or a paste.
Still soft or tacky after a day	Too much tint or an uneven mix. Scrape it out, clean with alcohol, redo with less tint and a longer mix.
Ragged edge after peeling	Shave the edge with the razor when leathery; peel sooner next time.
A low spot or pinhole after curing	Scuff it, tape round it, and fill just that spot with a small fresh batch.

Glow (optional, matches Drum Mark II)

Glow powder mixed into the yellow fill makes the lines glow faintly in a dim room. In daylight it only makes the yellow a little paler. The grooves are only 1mm deep, so the glow is modest. The powder is gritty and settles: fold it in gently and stir the blob again every few minutes while filling.

Safety: nitrile gloves, good ventilation, and keep epoxy off your skin. Change razor blades often and cut away from your hand. Let rags and leftover epoxy cure before throwing them away.